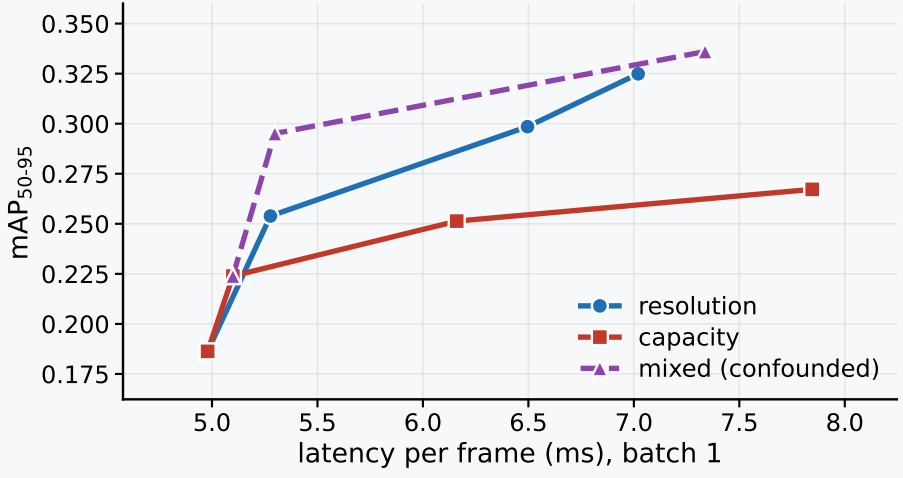
# Where should a fixed, measured inference budget go: resolution or capacity?

Real-time UAV detection, with latency and energy measured on the device at batch one rather than inferred from parameters or FLOPs

## Accuracy against measured cost




**resolution +0.0386**
at equal latency, 5.07–6.95 ms

**resolution +0.0550**
at equal energy, 0.29–0.76 J

With the device's idle draw removed, resolution leads by +0.0582 over 0.10–0.46 J.

## Without any interpolation

A configuration dominates another when it is both more accurate and cheaper: no grid, no curve fitting, no model.

**n@1536 ≻ l@640**

| | |
|---|---|
| accuracy | **+0.0577 mAP** |
| latency | **−10.5%** |
| energy | **−38.5%** |
| energy above idle | **−49.8%** |

15 such pairs on VisDrone, 15 of them in every seed; this one holds in all 3 seeds separately.

**No capacity configuration dominates a resolution one, on any axis.**

## The corrected experiment

An arm built as one model at several input sizes varies both factors. It leads the isolated arm by +0.0666 on latency and reverses the ordering against resolution. Both versions are reported.

## Three eliminations came first

Data enrichment: no gain over 60 controlled runs.
Confidence correction: low confidence on small objects is accurate, not a bias.
Our own first hypothesis: rejected once cost was measured instead of assumed.

## Second dataset

On UAVDT the direction replicates at a smaller magnitude, +0.0131 on energy over 0.28–0.43 J, and nothing reaches corrected significance. A check on the direction of the result, not a confirmation of it.